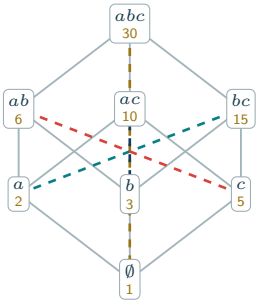
COMBINATORICS ON ONE PAGE

Partitions, cubes and Stirling numbers

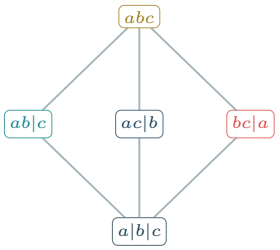
A subset is a corner of a cube, a partition is a different way to cut the same things, and the Stirling numbers count the cuts.

1 / THE RULE AND THE PICTURE

Antipodal pairs of the cube are the partitions into at most two blocks



subsets of $\{a, b, c\}$, gold: divisors of 30



5 partitions, $1, 3, 1$ with $1, 2, 3$ blocks

A **partition** cuts a set into non-empty blocks that do not overlap and cover everything, with no order on the blocks or on the elements inside them, so $ab\,|\,c$ is one partition of $\{a, b, c\}$. A *subset* is a different cut: choosing $S \subseteq \{a, b, c\}$ splits the set into S and its complement, and either part may be empty. The 8 subsets are the corners of a cube, and complementing sends each corner to the antipodal one. The 4 antipodal pairs are exactly the ways to cut $\{a, b, c\}$ into at most two blocks, drawn dashed:

$$\#\{\text{cuts into} \le 2 \text{ blocks}\} = S(n,1) + S(n,2) = 2^{n-1}$$

With $a, b, c = 2, 3, 5$ the pair $(d, 30/d)$ is a factorisation of 30.

2 / THE CUBOID OF DIVISORS

Divisibility makes the divisors of a number into a box, and the cube is the case of distinct primes

The divisors of N ordered by "divides" form a category with one arrow $d \to e$ whenever $d \mid e$, and it is a product of chains, one for each prime. A prime that occurs e times gives a chain of $e + 1$ steps, so N gives a box with $\prod(e_i + 1)$ corners:

number	prime factorisation	shape of the divisor box	corners
12	$2^2 \cdot 3$	3×2	6
30	$2 \cdot 3 \cdot 5$	$2 \times 2 \times 2$, a cube	8
60	$2^2 \cdot 3 \cdot 5$	$3 \times 2 \times 2$	12
210	$2 \cdot 3 \cdot 5 \cdot 7$	$2 \times 2 \times 2 \times 2$	16
360	$2^3 \cdot 3^2 \cdot 5$	$4 \times 3 \times 2$	24

For $30 = 2 \cdot 3 \cdot 5$ the three chains have two corners each, which is the cube of subsets of $\{2, 3, 5\}$. Replacing the primes by $2, 3, 7$ gives 42, or by three kinds of privilege gives the cube of your photographs: the shape of the relationships stays and the objects are forgotten, which is the extra level of abstraction. Arranging the divisors by size instead of by prime skews the cube into a cuboid.

3 / STIRLING NUMBERS OF THE SECOND KIND

$S(n,k)$ counts the partitions of an n-set into k blocks

Look at the last element n: it either joins one of the k blocks of a partition of the other $n-1$ elements, or stands alone as a new block, so

$$S(n,k) = k\,S(n-1,k) + S(n-1,k-1)$$

and the rows sum to the Bell numbers B_n, the number of partitions of any kind:

n	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	Bell B_n
1	1						1
2	1	1					2
3	1	3	1				5
4	1	7	6	1			15
5	1	15	25	10	1		52
6	1	31	90	65	15	1	203

The cube explains the second column. Sending each element to 0 or 1 gives 2^n cuts, of which 2 leave a block empty, and swapping the two labels pairs up the rest, so $S(n,2) = \frac{2^n - 2}{2} = 2^{n-1} - 1$. For $n = 4$ this gives 7, the four pairs $x\,|\,\text{rest}$ and the three pairs $xy\,|\,zw$. In general $k!\,S(n,k) = \sum_j (-1)^{k-j} \binom{k}{j} j^n$ counts the ways to map onto k labelled boxes.

4 / FACTORISATIONS AND THE SIZE OF THE CUBE

The same numbers count the ways to factor a number with distinct primes

An unordered factorisation of a squarefree number into factors above 1 is a partition of its set of primes, so it is counted by $S(n,k)$ with k factors. For 30 the row $1, 3, 1$ is 30; $2 \cdot 15$, $3 \cdot 10$, $5 \cdot 6$; $2 \cdot 3 \cdot 5$. The number 42 has the same shape with $2 \cdot 21$, $3 \cdot 14$, $6 \cdot 7$, and $210 = 2 \cdot 3 \cdot 5 \cdot 7$ has $1 + 7 + 6 + 1 = 15$. Partitions soon outgrow the cube of subsets, because B_n passes 2^n at $n = 5$:

	$n=1$	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$
cube 2^n	2	4	8	16	32	64	128	256
Bell B_n	1	2	5	15	52	203	877	4140

ASIDE / TRY THESE

Count $S(5,2) = 2^4 - 1 = 15$ and check it for $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$, which has 52 unordered factorisations. The first Stirling numbers $c(3,k) = 2, 3, 1$ count permutations by cycles.

LINKS TO OTHER SHEETS

The divisor cube is the product of chains from the cuboid idea; the polygon sheets use another lattice, the roots of unity.

Convention. $S(n,k)$ second-kind Stirling number; $B_n = \sum_k S(n,k)$ the Bell number; $ab\,|\,c$ lists the blocks of a partition; counts were checked by direct enumeration.

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